

VoxAI

Complete SaaS Product Documentation

AI-powered medical voice consultation platform. Doctors speak during patient consultations and VoxAI automatically transcribes the audio, extracts structured medical data (diagnoses, symptoms, medications), generates professional PDF reports, and delivers them via email & WhatsApp — all in real-time.

Property	Value
Product Name	VoxAI
Version	2.0.0
Company	AutomaticXAI (automaticxai.online)
Document Date	February 20, 2026
Classification	Confidential

Table of Contents

1. Prerequisites & Installation
2. Technology Stack
3. Architecture Overview
4. Database Schema
5. Backend API Endpoints
6. Frontend Pages & Components
7. Project Directory Structure
8. Environment Variables
9. User Roles & Access Control
10. Core Consultation Workflow
11. Pending Work & Issues
12. External Dependencies Summary

1. Prerequisites & Installation

What You Need to Install

Requirement	Purpose	Notes
Python 3.10+	Backend runtime	Required for Flask server
Node.js 18+	Frontend build tool	Required for Vite dev server
FFmpeg	Audio conversion	Required by pydub for Whisper preprocessing
Gemini API Key	AI medical extraction	Free tier from Google AI Studio
Supabase Account	Cloud database & auth	Free tier available

■■ **IMPORTANT:** Ollama is NOT used in this project. VoxAI uses **faster-whisper** (runs locally, no API needed) for transcription and **Google Gemini 2.0 Flash Lite** (cloud API) for medical data extraction.

Quick Start — Backend

```
cd backend
pip install -r requirements.txt
python app.py      # Starts on http://localhost:5001
```

Quick Start — Frontend

```
cd frontend
npm install
npm run dev        # Starts on http://localhost:5173
```

2. Technology Stack

Frontend

Technology	Version	Purpose
React	18.2	UI library (JSX components)
Vite	5.0	Build tool & dev server
TailwindCSS	3.3	Utility-first CSS framework
Framer Motion	12.34	Animations & transitions
React Router DOM	7.13	Client-side routing (13 routes)
Axios	1.6	HTTP client for API calls
Socket.IO Client	4.8	Real-time WebSocket communication
Lucide React	0.294	Icon library
Three.js	0.183	3D graphics (Landing Page)
Radix UI	Latest	Accessible UI primitives
TypeScript	5.9	Type checking (dev dependency)

Backend

Technology	Version	Purpose
Flask	Latest	Python web framework
Flask-SocketIO	Latest	WebSocket support for live audio
Flask-CORS	Latest	Cross-origin request handling
faster-whisper	Latest	Local speech-to-text (OpenAI Whisper)
Google Generative AI	Latest	Gemini 2.0 Flash Lite — medical extraction
ReportLab	Latest	PDF report generation
Supabase Python	Latest	Database & storage client
pydub	Latest	Audio format conversion
python-dotenv	Latest	Environment variable management
requests	Latest	HTTP client (n8n webhooks)

Database & Infrastructure

Service	Purpose
Supabase (PostgreSQL)	Primary database, authentication, file storage
Supabase Auth	User authentication with Row Level Security
Supabase Storage	Audio file & PDF storage
n8n (self-hosted on Hostinger)	Workflow automation — email & WhatsApp notifications

AI / ML Models

Model	Provider	Purpose
Whisper (small)	OpenAI (via faster-whisper)	Speech-to-text — runs 100% locally on CPU
Gemini 2.0 Flash Lite	Google AI	Medical data extraction from transcriptions (cloud API)

3. Architecture Overview

VoxAI follows a **decoupled client-server architecture** with a React SPA frontend communicating with a Flask REST/WebSocket backend. The backend orchestrates two AI pipelines (Whisper for STT, Gemini for medical extraction) and persists all data to Supabase (PostgreSQL). Notifications are dispatched via n8n webhooks hosted on Hostinger.

Data Flow

Step	Component	Description
1	Frontend (React)	Doctor enters patient info, records or uploads audio
2	Backend (Flask)	Receives audio via REST API or WebSocket
3	Whisper AI (Local)	Transcribes audio to text (runs on CPU, no API key)
4	Gemini 2.0 (Cloud)	Extracts structured medical data from transcription
5	Supabase (Cloud)	Stores consultation record (transcription + medical JSON)
6	Doctor Review	Doctor reviews, edits, and approves the AI output
7	PDF Generator	ReportLab generates professional medical report PDF
8	n8n Webhook	Sends PDF via email & WhatsApp to patient

4. Database Schema (Supabase / PostgreSQL)

Tables

Table	Purpose	Key Fields
clinics	Multi-tenant clinic orgs	id, name, location, plan, logo_url, header/footer_template
profiles	Users (doctors & admins)	id (FK→auth.users), clinic_id, full_name, email, role, photo_url
patients	Patient records per clinic	id, clinic_id, name, phone_number, email, age
consultations	Consultation records	id, clinic_id, doctor_id, patient_id, transcription, medical_data (JSONB)
demo_requests	Lead capture (landing page)	id, full_name, professional_email, clinic_name, status
interest_submissions	Onboarding interest forms	id, clinic_name, professional_email, description, status

Row Level Security (RLS)

- **Profiles:** Users see only their own data
- **Patients:** Scoped to user's clinic_id
- **Consultations:** Scoped to user's clinic_id
- **Demo Requests:** Public insert, admin-only read
- **Interest Submissions:** Public insert, admin-only read

5. Backend API Endpoints

Core Consultation Flow

Method	Endpoint	Description
POST	/api/transcribe	Upload audio → Whisper → Gemini → save to Supabase
POST	/api/upload-audio	Upload existing audio file for processing
POST	/api/upload-transcript	Submit raw text transcript (skip Whisper)
POST	/api/approve	Approve → generate PDF → send via n8n
GET	/api/patient-history/<phone>	Retrieve all past consultations by phone

Authentication & Profiles

Method	Endpoint	Description
POST	/api/login	Integrated login for doctors & clinic admins
POST	/api/complete-registration	Doctor sets name & password after invite
POST	/api/upload-profile-photo	Upload/update profile photo
PUT	/api/update-profile	Update profile details

Clinic & Staff Management

Method	Endpoint	Description
GET	/api/clinics	Master admin: list all clinics
POST	/api/clinics	Master admin: create new clinic
PUT	/api/clinics/<id>	Master admin: update clinic
DELETE	/api/clinics/<id>	Master admin: delete clinic
GET	/api/clinic-staff	Clinic admin: list doctors
POST	/api/invite-doctor	Clinic admin: invite a doctor
DELETE	/api/clinic-staff/<id>	Clinic admin: remove a doctor
POST	/api/generate-invite-link	Generate invite link for doctor

Analytics, History & Branding

Method	Endpoint	Description
GET	/api/doctor-history/<id>	Consultations for a specific doctor
GET	/api/clinic-history	All consultations for a clinic
GET	/api/daily-stats	Daily activity statistics
GET	/api/clinic-branding/<id>	Get clinic branding settings
PUT	/api/clinic-branding	Update clinic branding

Lead Capture & WebSocket Events

Method/Event	Endpoint	Description
POST	/api/demo-request	Submit demo request from landing page
POST	/api/submit-interest	Submit interest form
WS: connect	—	Client connected via WebSocket
WS: disconnect	—	Client disconnected
WS: audio_chunk	—	Live audio chunk for real-time transcription

6. Frontend Pages & Components

Pages (13 Routes)

Route	Page	Description
/	LandingPage	Marketing page with demo request, interest forms, 3D effects
/login	LoginPage	Unified login for doctors & admins
/register	RegisterPage	Doctor registration via invite link
/dashboard	Dashboard	Doctor workspace — consultation wizard, history, profile
/admin	AdminPanel	Master admin — clinic management, staff, analytics
/about	AboutPage	About the company
/blog	BlogPage	Blog articles
/partners	PartnersPage	Partner information (automaticxai.online)
/privacy	PrivacyPage	Privacy policy
/terms	TermsPage	Terms of service
/hipaa	HIPAAPage	HIPAA compliance details
/security	SecurityPage	Security practices
/faq	FAQPage	Frequently asked questions

Key Components

Component	Size	Purpose
ConsultationWizard	54 KB	Multi-step consultation flow (patient → record → review → approve)
UploadWizard	48 KB	File upload-based consultation flow
Dashboard	38 KB	Main doctor dashboard (tabs, history, stats)
AdminPanel	36 KB	Master admin panel (clinic CRUD, staff, analytics)
TeamManagement	18 KB	Clinic admin doctor management
ProfileSettings	16 KB	User profile editing
BrandingSettings	14 KB	Clinic branding customization
DoctorApprovalScreen	13 KB	Review & approve AI-generated medical report
LiveRecorder	10 KB	Real-time audio recording via WebSocket
DemoRequestModal	9 KB	Landing page demo request form

Component	Size	Purpose
FileUpload	8 KB	Audio file upload component
PatientInfoForm	5 KB	Patient data input form
MedicationCard	2.5 KB	Individual medication display card
ReportViewer	1.7 KB	Medical report display
TranscriptionDisplay	1.9 KB	Transcription text viewer
ThemeProvider / ThemeToggle	3 KB	Dark/light theme support

7. Project Directory Structure

```
VoxAI/
├── backend/
│   ├── app.py                # Main Flask app (1137 lines, 30+ endpoints)
│   ├── config.py             # Configuration from .env
│   ├── .env                  # Environment variables
│   ├── requirements.txt       # Python dependencies
│   ├── models/
│   │   ├── transcription.py   # Whisper AI speech-to-text
│   │   ├── extraction.py      # Gemini medical data extraction
│   │   └── services/
│   │       ├── pdf_generator.py # ReportLab PDF generation
│   │       ├── report_generator.py # Structured report builder
│   │       └── notification_service.py # n8n webhook integration
│   ├── uploads/              # Temporary audio uploads
│   └── pdfs/                  # Generated PDF reports
│
├── frontend/
│   ├── package.json           # Node.js dependencies
│   ├── vite.config.js         # Vite configuration
│   ├── tailwind.config.js     # TailwindCSS config
│   ├── index.html             # SPA entry point
│   └── src/
│       ├── App.jsx            # Route definitions
│       ├── main.jsx           # React entry point
│       ├── pages/              # 13 page components
│       ├── components/        # 16 feature components
│       │   └── ui/             # 5 reusable UI primitives
│       ├── constants/         # App constants
│       ├── lib/               # Utility functions
│       └── assets/            # Static assets
│
└── supabase_schema.sql        # Database schema definition
```

8. Environment Variables

Variable	Description	Example
FLASK_ENV	Flask environment	development
SECRET_KEY	Flask secret key	your-secret-key
WHISPER_MODEL	Whisper model size	small (tiny/base/small/medium/large)
SUPABASE_URL	Supabase project URL	https://xxx.supabase.co
SUPABASE_KEY	Supabase anon key	eyJhbG...
GEMINI_API_KEY	Google AI API key	AlzaSy...
N8N_WEBHOOK_URL	n8n webhook endpoint	https://your-n8n.hostinger.com/webhook/...
MAX_FILE_SIZE	Max upload size (bytes)	104857600 (100 MB)

9. User Roles & Access Control

Role	Access	Created By
Master Admin	Full system — create/delete clinics, manage all staff, view analytics	System owner
Clinic Admin	Manage their clinic's doctors, view clinic stats, invite doctors	Master Admin
Doctor	Run consultations, view own history, manage profile	Clinic Admin (via invite)

Onboarding Flow

1. **Master Admin** creates a new clinic via the Admin Panel
2. **Master Admin** or **Clinic Admin** generates an invite link
3. **Doctor** receives link → registers with name & password
4. **Doctor** logs in → accesses Dashboard → runs consultations

10. Core Consultation Workflow

The following describes the end-to-end flow of a single patient consultation through VoxAI:

Step	Direction	Description
1	Doctor → Frontend	Doctor enters patient info (name, phone, age)
2	Doctor → Frontend	Doctor records audio live or uploads an audio file
3	Frontend → Backend	POST /api/transcribe with audio file
4	Backend → Whisper	Audio converted to 16kHz mono WAV, transcribed locally
5	Backend → Gemini	Transcription sent to Gemini 2.0 Flash Lite for medical extraction
6	Gemini → Backend	Returns structured JSON: diagnoses, symptoms, medications, recommendations
7	Backend → Supabase	Consultation record saved (transcription + medical_data JSONB)
8	Backend → Frontend	Returns full result to doctor
9	Doctor → Frontend	Reviews, edits medications/diagnoses, approves report
10	Frontend → Backend	POST /api/approve with final data
11	Backend → ReportLab	Generates professional PDF medical report
12	Backend → n8n	Sends PDF + patient info to n8n webhook
13	n8n → Patient	Delivers report via email and/or WhatsApp

11. Pending Work & Issues

■ Critical (Must Fix)

#	Item	Details
1	n8n Webhook Not Configured	The .env still has a placeholder URL. Email & WhatsApp delivery is non-functional.
2	No Authentication Middleware	API endpoints have no JWT/token validation. Any client can call any endpoint.
3	Passwords Stored in Plaintext	The profiles table stores passwords as plain text. Must use bcrypt/argon2.
4	API Keys Exposed in .env	Supabase key and Gemini API key are committed to the repo. Must rotate & gitignore.
5	No Route Guards (Frontend)	Dashboard and Admin routes are not protected — accessible without login.

■ Important (Should Build)

#	Item	Details
6	WebSocket Live Transcription	audio_chunk handler is a stub — no real-time transcription during recording.
7	Supabase Storage Integration	Files saved locally to uploads/ and pdfs/ — not uploaded to Supabase Storage.
8	Production Deployment	No Dockerfile, docker-compose, CI/CD pipeline, or deployment config.
9	Error Handling & Logging	Uses print() instead of proper Python logging. No Sentry/error tracking.
10	Rate Limiting	No rate limiting on any endpoints — vulnerable to abuse.
11	Gemini Quota Management	No retry logic, fallback model, or queue for high-volume usage.
12	Multi-Language UI	Whisper supports 99 languages but UI is English-only. No i18n framework.

■ Nice to Have (Future Enhancements)

#	Item	Details
13	Unit & Integration Tests	No test suite. One-off scripts exist but no pytest/jest framework.
14	PDF Template Customization	Clinic header/footer templates not fully integrated into PDF generator.
15	Patient Portal	No patient-facing interface. Reports only via email/WhatsApp.

#	Item	Details
1 6	Advanced Analytics	Limited analytics. Need trends, charts, doctor performance dashboards.
1 7	Offline Mode (PWA)	No service worker for offline use in low-connectivity areas.
1 8	API Documentation	No Swagger/OpenAPI spec for the 30+ endpoints.
1 9	Billing & Payments	plan field exists but no Stripe/Razorpay integration.
2 0	Audit Logging	No audit trail for patient data access (important for HIPAA).

12. External Dependencies Summary

Dependency	Type	Required?
FFmpeg	System binary	■ Yes — required by pydub for audio conversion
Google Gemini API	Cloud API	■ Yes — medical extraction fails without it
Supabase	Cloud service	■ Yes — all data storage and auth
n8n on Hostinger	Self-hosted	■■ Optional — only for email/WhatsApp delivery
Ollama	—	■ NOT USED — project uses faster-whisper + Gemini

End of Documentation — VoxAI v2.0.0